

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CS41714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (B1A)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
Total Marks: 30

QUESTIONS: 5

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Tharun (192412004) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Bis*

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RUBRICS FOR CALCULATION

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps, multiple errors	No clear process
Accuracy of Calculations	20%	All calculations accurate, correct final answer	Minor calculation errors	Frequent errors, incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Some interpretation with minor omissions	Limited or no interpretation	No interpretation

Medical Attention Alert

Propositional Logic

Let:

* F = Ravi has a fever

* M = Ravi needs medical attention

Knowledge Base:

1. If Ravi has a fever, then Ravi needs medical attention

$$F \rightarrow M$$

2. Ravi has a fever

F

Goal:

M

2. CNF Conversion and Negation of Goal

Convert the implication:

$$F \rightarrow M = \neg F \vee M$$

CNF

clauses:

$$* C_1 = \neg F \vee M$$

$$* C_2 = F$$

Negate the goal:

$$\neg M$$

So, the clauses are:

$$\{\neg F \vee M, F, \neg M\}$$

3. Resolution steps

Resolve C_1 and C_2 :

$$(\neg F \vee M), F$$

Resolving F and $\neg F$:

$$C_3 = M$$

Now resolve C_3 with the negated goal:

$$M, \neg M$$

Therefore:



4. Conclusion:

Since the empty clause is obtained, the goal is proved.

Ravi needs medical attention

2. Library Late Fine

1. Propositional Logic

Let:

* L = The book was returned late.

* F = A fine is charged.

Knowledge Base:

1. If a book is returned late, then a fine is charged.

$$L \rightarrow F$$

and:

2. The assumption and negation of and
cannot be resolved:

$$L \rightarrow F \equiv \neg L \vee F$$

OK done

$$\# L = \neg L \vee F$$

$$\# C_1 = L$$

Accepted goal:

$$\neg F$$

Therefore:

$$\{\neg L \vee F, L, \neg F\}$$

3. Resolution steps:

Resolve C_1 & C_2 :

$$(L \vee F), L$$

Resolving L and $\neg L$:

$$C_3 = F$$

Now resolve C_3 with the accepted goal:

$$F, \neg F$$

Therefore:



4. Conclusion:

The empty clause is obtained, so the goal is proved.

A fire is charged.

3. Flight Delay Notification

1. Propositional Logic

Let:

* D : Flight 202 is delayed

* N : Passengers are notified.

Knowledge Base:

1. If the flight is delayed, then passengers are notified.

$$D \rightarrow N$$

2. Flight 202 is delayed.

D

Goal:

N

2. CNF conversion and Negation of goal

convert implications:

$$D \rightarrow N \equiv \neg D \vee N$$

WF clauses:

$$* C_1 = \neg D \vee N$$

$$* C_2 = D$$

Negate the goal:

$$\neg N$$

Resolve C_1 and C_2 :

$$(\neg D \vee N), D$$

Resolving D and $\neg D$:

$$C_3 = N$$

Now resolve C_3 with the negated goal:

$$N, \neg N$$

Therefore:



4. Conclusion.

Since the empty clause is derived, the goal is proved.

Passengers are notified.

4. Employee Certification

1. Propositional Logic:

Let:

* T = Kevin completed the training

* C = Kevin receives certification

Knowledge Base:

1. If any employee completes the training then the employee receives certification

$$T \rightarrow C$$

2. Kevin completed the Training T

Goal:

C

2. CNF conversion and Negation of goal:

convert implication:

$$T \rightarrow C \equiv \neg T \vee C$$

CNF clauses:

$$* C_1 = \neg T \vee C$$

$$* C_2 = T$$

Negated goal:

$$\neg C$$

Therefore:

$$\{\neg T \vee C, T, \neg C\}$$

3. Resolution steps:

Resolve C_1 and C_2 :

$$(\neg T \vee C), T$$

Resolving $T \wedge \neg T$:

$$C_3 = C$$

Now resolve C_3 with the negated goal:

$$C, \neg C$$

Therefore:



4. conclusion

The empty clause is obtained. Hence the goal is proved

Kevin Newman a certification

- * R : The car is certified reading.
- * h : The car can be registered.

Knowledge Base.

1. If the car passes the entrance test, then it is certified reading.
 $E \rightarrow R$
2. If the car is certified reading, then it can be registered.
 $R \rightarrow h$
3. The car passed the entrance test.

Goal:

h

2. CNF conversion.

Convert the first implication:

$$E \rightarrow R$$

$$\boxed{\neg E \vee R}$$

Convert the second implication:

$$R \rightarrow h$$

$$\boxed{\neg R \vee h}$$

Fact:

$$\boxed{E}$$

Therefore the CNF clauses are:

$$* C_1 = \neg E \vee R$$

$$* C_2 = \neg R \vee h$$

$$* C_3 = E$$

3. Negate the goal:

Goal:

G

Negated goal:

$\neg G$

Complete set of clauses:

$\{\neg E \vee R, \neg R \vee G, E, \neg G\}$

4. Resolution Algorithm

Step 1: Resolve $C_1 \wedge C_3$

$(\neg E \vee R), E$

Resolve $E \wedge \neg E$:

$C_4 = R$

This means the car is certified roadworthy

Step 2: Resolve $C_2 \wedge C_4$

$(\neg R \vee G), R$

Resolve $R \wedge \neg R$:

$C_5 = G$

This means the car can be registered

Step 3: Resolve C_5 with the negated goal

$G, \neg G$

Resolving $G \wedge \neg G$:

$\square$

5. Final conclusion

Since the empty clause is obtained, the negation of the goal leads to a contradiction

Therefore, the original goal is true:

The car can be registered.